exploration, the token we design stands for numerous other assets in many businesses, as well as in socioeconomic, cultural, and art applications.

We'll begin with a problem statement and then apply the design principles (appendix B) to design and develop the application. This real estate token will be called RES4 (real estate for all).

> **PROBLEM STATEMENT** Design and develop a real estate token decentralized application representing new real estate developments in a town. The town supervisor can add a piece of real estate as an asset (RES4 token) and at the same time assign it to an owner. This task is accomplished by the process of the creation of the RES4 token. (Assume that the funds for asset ownership are transferred by other means that are not within the scope of this problem.) The owner of the token can add value to the token by building on it, as well as approve a sale to a buyer, and an approved buyer can buy the asset. The real estate asset may also appreciate or depreciate as determined by a town's assessor. For simplicity, assume that the town supervisor and the assessor have the same identity, and that they represent the town and perform operations on behalf of the town.

RES4 is a simplified version of the real-world real estate business. You can improve this basic design after you complete development of the RES4 smart contract and Dapp.

9.2.1 Use case diagram

To get to the smart contract design, you begin with design principle 2: designing the use case diagram. Figure 9.3 shows the actors for the RES4 token as follows:

- Town supervisor (developer or creator of the asset)
- Owner of the asset
- Builder of the asset (adds value to asset)
- Buyer of asset
- Assessor of the value of the asset